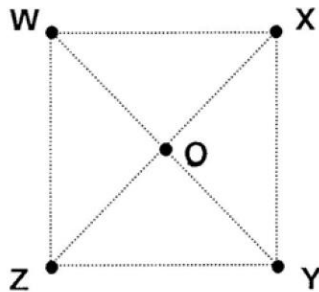


resistor is going to be $\frac{1}{4}$ that of X. Thus power is 2.5 W.

- 24** The diagram shows four long straight current-carrying conductors placed at the corners of a square at W, X, Y and Z. The current in each conductor has the same magnitude.



Point O is the intersection of the diagonals of the square. If the magnetic flux density at point O is zero, which of the following is true?

- A** The current in Y must be in the opposite direction as that in W and the current in X must be in the opposite direction as that in Z.
- B** The current in Y must be in the same direction as that in W and the current in X must be in the opposite direction as that in Z.
- C** The current in Y must be in the same direction as that in X and the current in W must be in the opposite direction as that in Z.
- D** The current in Y must be in the same direction as that in X and the current in W must be in the same direction as that in Z.

Ans: D

If I_Y and I_X are in the same direction, they will create B_Y and B_X to produce a net B either upwards or downwards. Therefore, I_W and I_Z must also be in the same direction as I_Y and I_X to produce no net B at point O. $\rightarrow$ i.e. current in all four wires must be in the same direction.